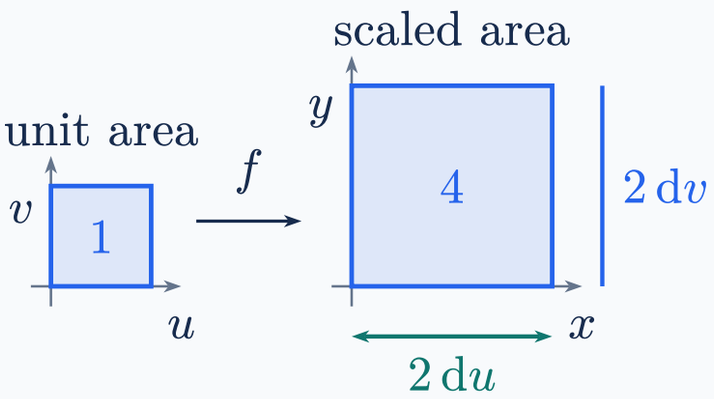
Change of variables

The Jacobian measures how a tiny coordinate cell changes area or volume.

$$\int_{f(U)} g(\boldsymbol{x})\,\mathrm{d}\boldsymbol{x} = \int_{U} g(f(\boldsymbol{u}))\left|\det J_f(\boldsymbol{u})\right|\,\mathrm{d}\boldsymbol{u}$$

Linear scaling

$$x = 2u, \quad y = 2v$$



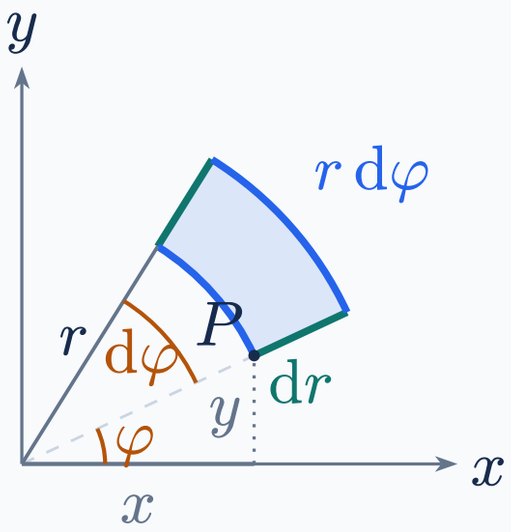
$$2 \times 2 = \left|\det J_f\right| = 4$$

$$\mathrm{d}A = 4\,\mathrm{d}u\,\mathrm{d}v$$

Each side doubles, so area grows by four. Reflecting one axis gives $\det J_f = -4$; the area factor stays 4.

Polar coordinates

$$x = r\cos\varphi, \quad y = r\sin\varphi$$



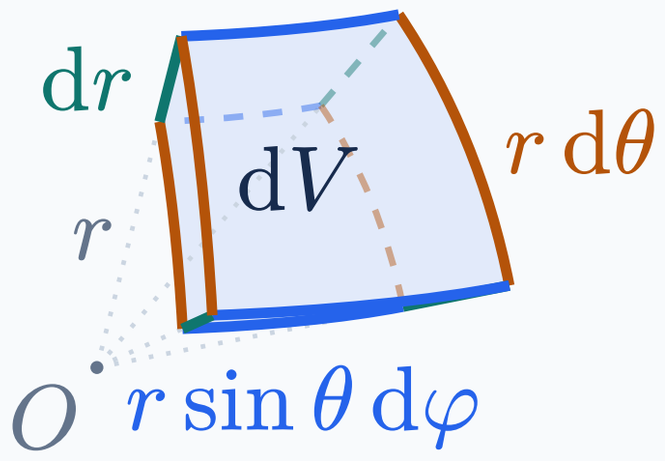
$$1 \times r = \left|\det J_f\right| = r$$

$$\mathrm{d}A = r\,\mathrm{d}r\,\mathrm{d}\varphi$$

The angular side is arc length $r\,\mathrm{d}\varphi$. Multiply it by the radial thickness $\mathrm{d}r$ to get the area element.

Spherical coordinates

$$x = r\sin\theta\cos\varphi$$
$$y = r\sin\theta\sin\varphi, \quad z = r\cos\theta$$



$$1 \times r \times r\sin\theta = r^2\sin\theta$$

$$\mathrm{d}V = r^2\sin\theta\,\mathrm{d}r\,\mathrm{d}\theta\,\mathrm{d}\varphi$$

Meridians have radius r; latitude circles have radius $r\sin\theta$. Their arc lengths supply the two angular factors.

Use the absolute determinant. Orientation can flip; an area or volume measure stays nonnegative. The map f is one-to-one on U.

Angles are in radians; θ is measured from $+z$. Cells are enlarged for clarity; labeled lengths describe the infinitesimal limit.